



香 港 大 學

THE UNIVERSITY OF HONG KONG

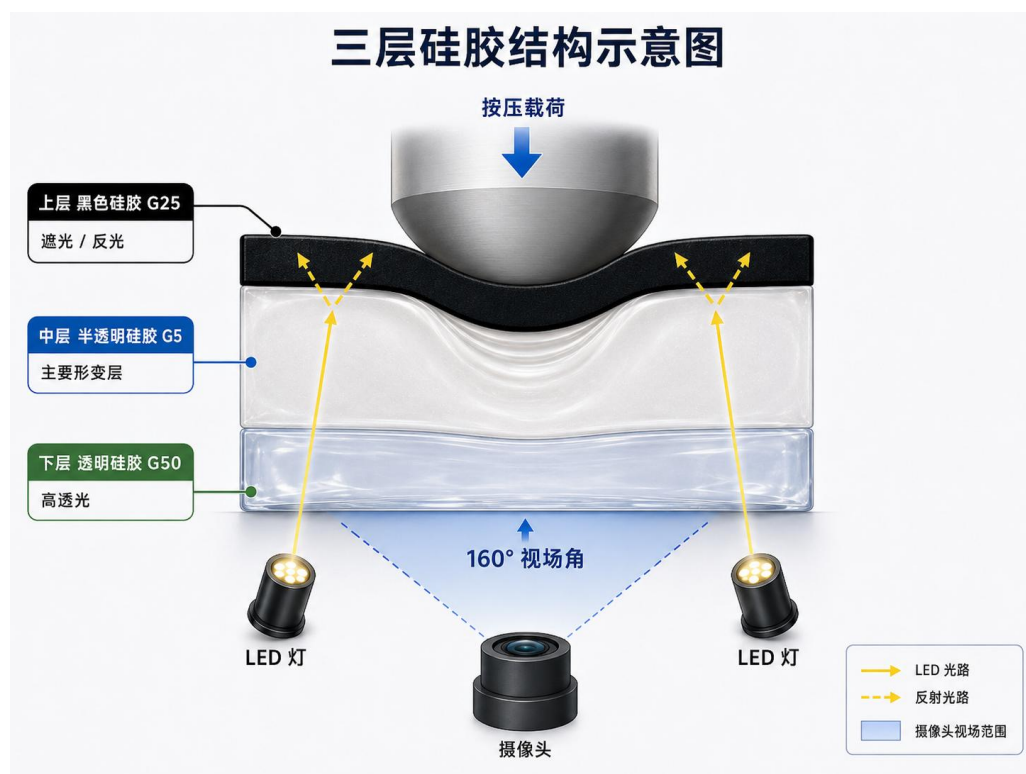
Progress Report

Name: Liu Zhaolin

Date: 19/07/2026

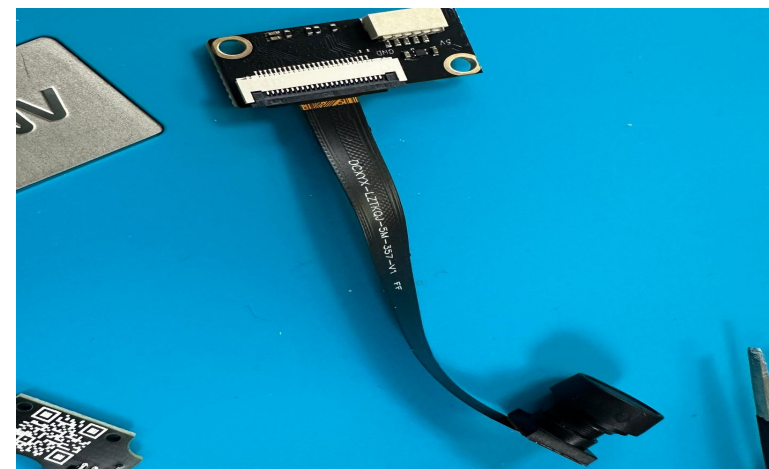
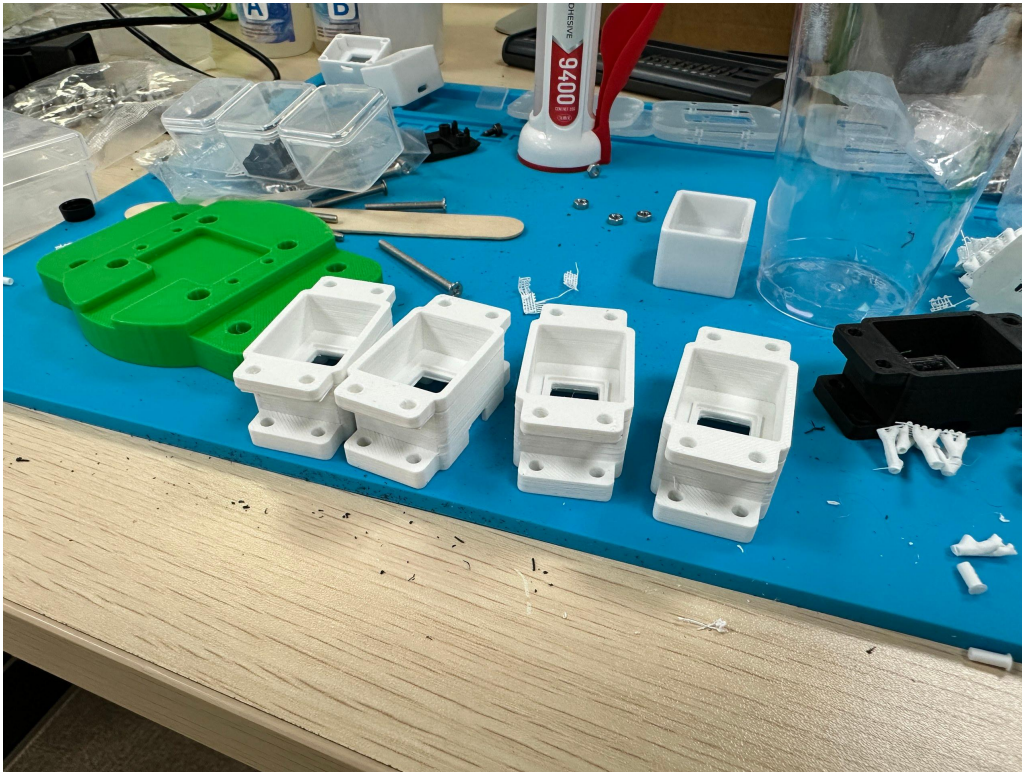
1.1 Hardware Setup of the Newly Designed Sensor

A completely new design was adopted to minimize the size of the tactile sensor while maximizing its measurement accuracy.



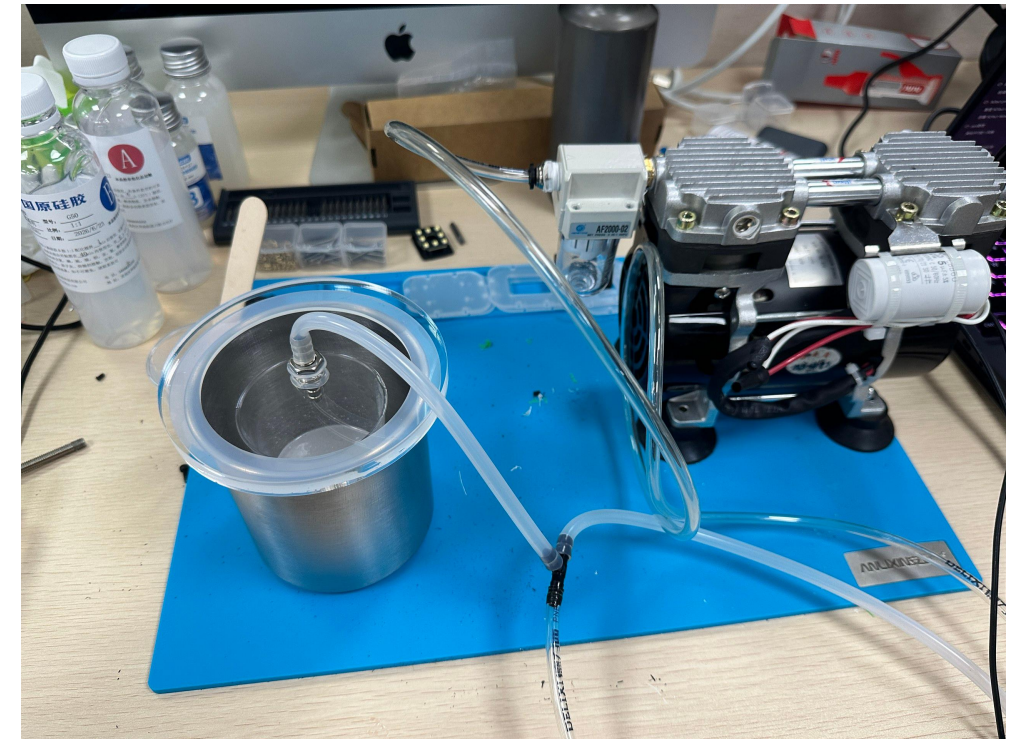
1.1 Hardware Setup of the Newly Designed Sensor

Hardware Fabrication Process



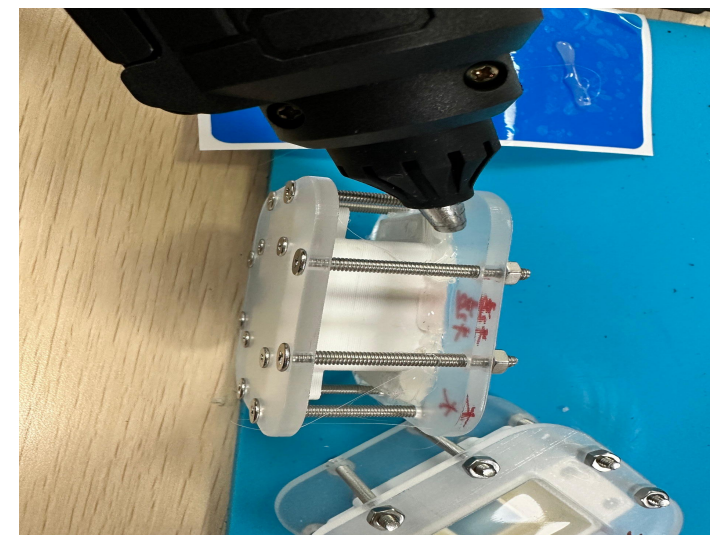
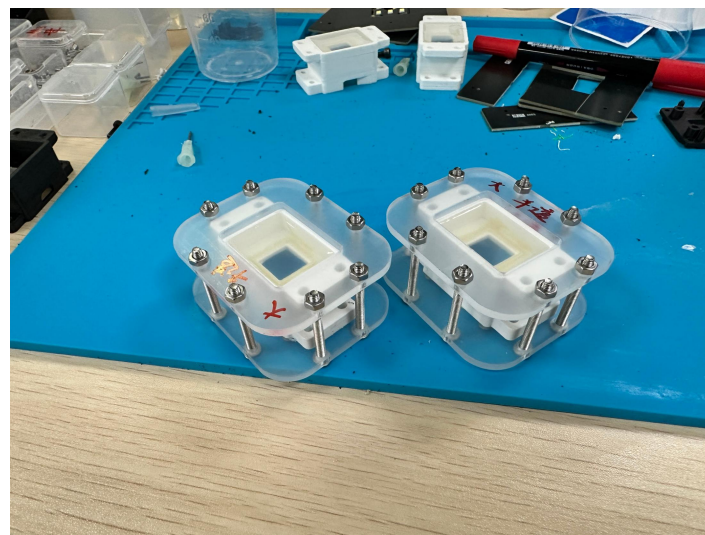
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Hardware Fabrication Process



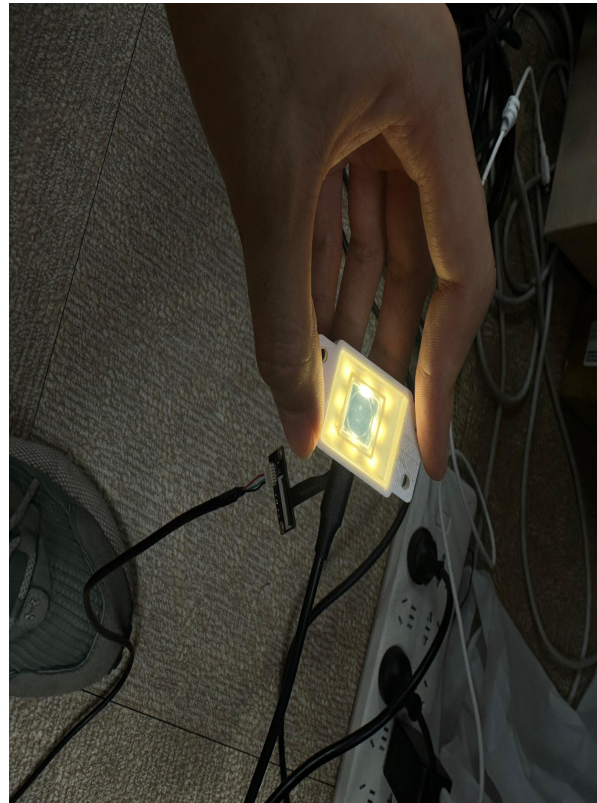
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Hardware Fabrication Process



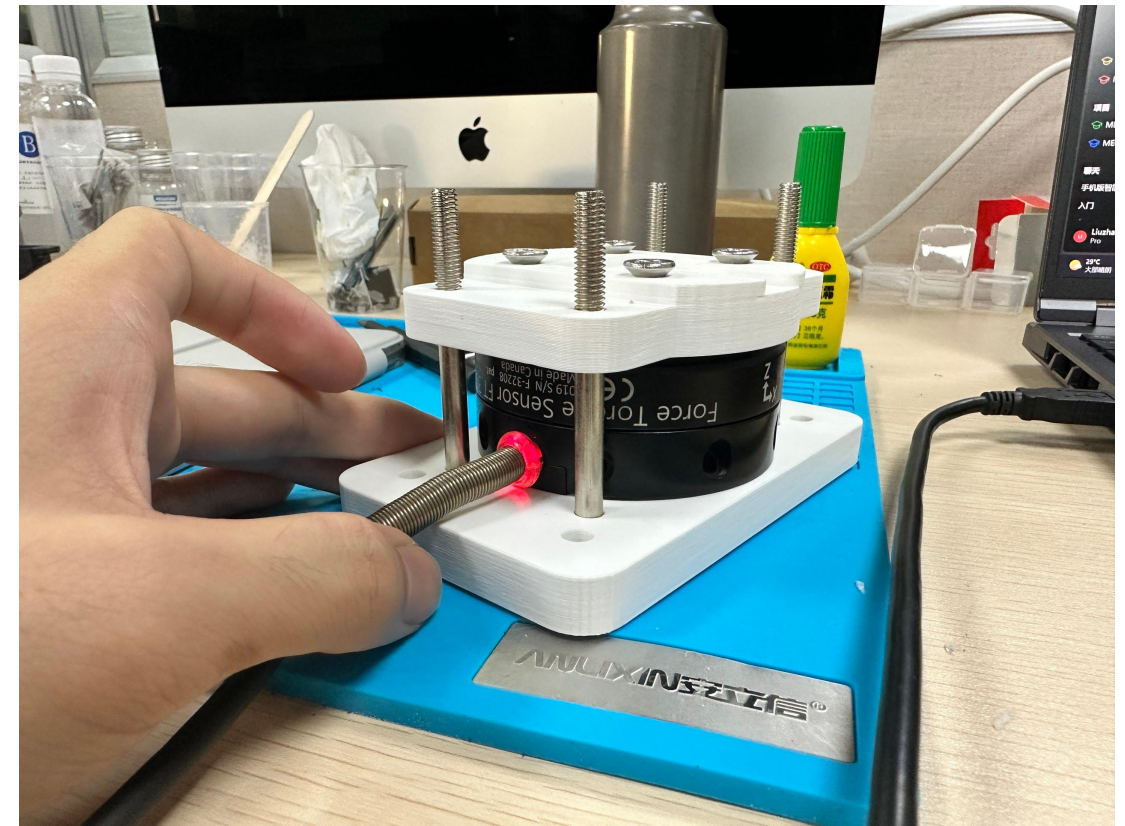
1.1 Hardware Setup of the Newly Designed Sensor

Electrical Testing



1.1 Hardware Setup of the Newly Designed Sensor

Six-Axis Force Sensor Setup and Testing



2. Next Week 's Work Plan

For the hardware:

- Build a synchronized acquisition system for 9DTact images and FT300 six-axis force labels.
- Collect a larger and more diverse dataset, including normal pressing, shear, eccentric loading, and torsion.
- Repeat experiments under different loads, directions, and contact positions to improve data reliability.
- Perform timestamp alignment, outlier removal, and data normalization.

For the software:

- Perform timestamp alignment, outlier removal, and data normalization.
- Train a model to estimate 6Dforces from 9Dtact images.
- Improve estimation accuracy through model tuning and additional data collection.
- Evaluate the model